

The Impact of Social Media on Mental Health, and what training models are effective.

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Abstract

This project seeks to examine the complex relationship between social media and its effect on mental health. While social media fosters connection, community, and self-expression, excessive use and exposure to idealized portrayals of life can lead to anxiety, depression, and isolation. By analyzing research and user insights, the project aims to better understand social media's impact on mental well-being and offer recommendations for healthier online engagement. The data we will be using contains features such as the platform used by the participant, their age, and daily social media activity statistics. These statistics include information such as the number of posts they make, comments they receive, and likes they garner. The models that are planned to be used are linear regression to predict user demographics based on daily usage, FNN neural network to identify relationships between dominant emotions and other features, k-means clustering to analyze patterns in social media usage, and random forest to classify participants based on their dominant emotion and usage patterns.

Introduction

Social media has become an inescapable part of life for the majority of people. Social media is a broad term that refers to platforms, applications, and other software that connects people all around the world. Many of these issues can end up affecting adolescents, showing a positive correlation between adolescent

depression and social media use[1]. Because of this various learning models needed to be deployed to evaluate the correlation between usage and the effect on emotions. Finding which models could be used to analyze certain factors the most effectively, and be able to properly detect when intervention might be needed is something valuable to users, guardians, and the future of society.

Though this is not the only issue, many others such as anxiety, negative self-image, anger, and various other feelings and emotions. While it might be natural to experience them, many of these can lead to a negative impact on one's life. Such that some have even considered using therapy-based interventions among adults, which has shown a positive correlation in improvement [2]. This is not something that can merely be overlooked, as it impacts everyone. Either directly or through someone close to them. Considering how certain platforms have a usage rate among teens, YouTube (93%), TikTok (63%), Snapchat(60%), and Instagram(59)[3]. While the amount that each teen uses can vary, the majority of the teens in this study visit each platform several times a day, and in the worst case constantly. While it is not all bad, it has been found that social media usage can lead to people feeling more connected, and interacting with a more diverse group, it can also lead to pressure on the view others might have on what they post, and drama on the platform [4]

However, finding these correlations is tough since improper diagnosis can also lead to

someone developing that same mental illness or others. It is then very important that there is a way to properly detect if someone has begun to experience these mental issues from their activity online with a high amount of accuracy. This is why certain models of training should be deployed, the highest accuracy will vary depending on the data being used. So being able to compare these different models, even when it is known why some might be ineffective for some sets of data. By finding what is effective, and most ineffective for what is being searched for within these diverse and vast data sets, it can help to improve the amount of successful detection done by a model. By using simplistic, and more advanced machine learning techniques, we used something as simple as linear regression and more advanced as a feed-forward neural network. The strengths, and weaknesses of these models, and understanding why and when to use them for this issue could lead to a large improvement in the issue we are focused on.

Related Works

Because of the complexity of trying to predict someone's emotions based on other factors, advanced and different machine learning models need to be deployed. Shown in the article *Detection of emotion by text analysis using machine learning* looks at how different models predict emotions based on text, showing high accuracy when using RNN.[5]

Along with that the article *A Hybrid CNN-LSTM Model for SMS Spam Detection in Arabic and English messages* was able to compare different machine learning models, from some we used in this such as Random Forest, and compare how well they performed at predicting spam texts [6]. In doing this they found that a combination of models was what performed best, which could

be an indicator as to how to properly solve the issue we are looking into. They also decided to focus on the accuracy of the model, given that they were evaluating text messages, while we are more focused on the amount [6]

The reason why accuracy is so important is that a misdiagnosis with something like this, as shown in the research paper *Diagnostic Error in Mental Health: a review*, depression had a 62% false positive rate, and the paper also goes into how many other mental health issues are being misdiagnosed[7]. It finds that many issues are made when diagnosing a patient, and because of the risk in doing that. It became important that when we analyzed our models we looked for high accuracy scores.

Proposed method

In this project, we implemented 4 models: linear regression, FNN, K-Means Clustering, and Linear Regression.

K-Means Clustering

K-Means clustering is a widely used unsupervised machine learning algorithm designed to identify patterns in a dataset by grouping similar data points into distinct clusters. Our data set contains a lot of user-level data, including but not limited to social media platforms, usage metrics (e.g. like received or time spent), and dominant emotion. K-means can find groupings between users that are more likely to be associated with specific emotional states. Additionally factors like one's social media platform and user related usage metrics (e.g. like received or time spent) can be focused on and create interpretable clusters that align with our project's direction and objective. We expect this model to give insight into behavioral and emotional patterns of social media users. These findings will primarily be for exploratory

purposes, and to assist our later research for a deep understanding on how social media and users' emotional state interact.

Random Forest

Random Forest is a versatile machine learning method that can be applied to a wide range of tasks, including classification and regression, depending on the dataset. In this project, Random Forest was utilized as a classification algorithm to predict the dominant emotion of users based on their social media usage patterns. The method works by building multiple decision trees during training, where each tree provides a prediction, and the final output is determined by majority voting across all trees. This ensemble approach helps to improve accuracy and reduce overfitting.

Given the complexity of our dataset, which included numerous metrics such as daily usage time, posts, likes, and comments, some models struggled with the multi-dimensional nature of the data. However, Random Forest effectively handled this by splitting the target variable into a multi-label classifier, enabling it to predict how each dominant emotion correlated with the input features. By focusing on dominant emotion as the target variable and aggregating predictions from multiple trees, Random Forest demonstrated its ability to handle diverse metrics and deliver reliable classification results.

Feedforward Neural Networks (FNN)

FNNs are a fundamental type of artificial neural network that consist of layers of interconnected nodes, where information flows in one direction—from the input layer to the output layer—without looping back. FNNs are designed for supervised learning tasks, such as

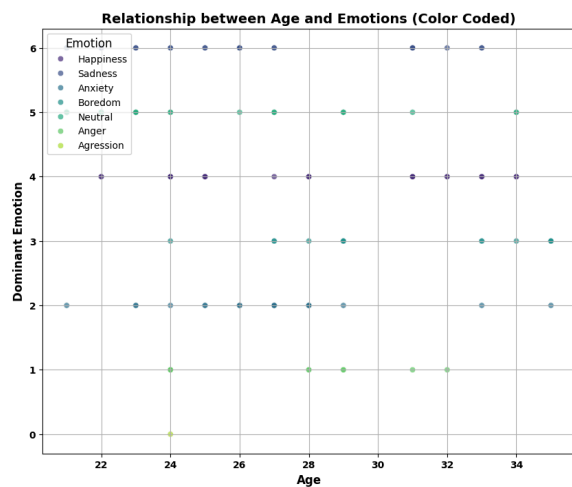
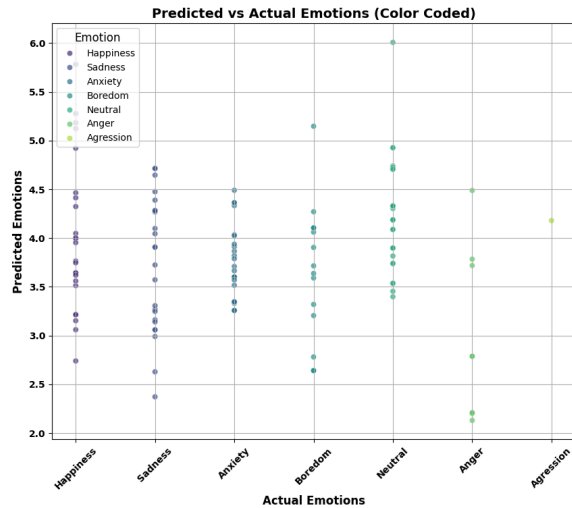
classification and regression, and excel at modeling non-linear relationships in data.

The key strength of FNNs lies in their ability to capture complex patterns through the use of multiple layers and non-linear activation functions. Each layer in an FNN applies a transformation to the input data, progressively learning abstract features. For optimization, FNNs rely on backpropagation to adjust weights and biases during training.

In this project, the FNN is used for multi-class classification to predict users' dominant emotions based on social media activity data.

Linear Regression

Linear regression was chosen as a foundational model for this project because of its simplicity, interpretability, and effectiveness in identifying baseline relationships between features and outcomes. As one of the most widely used statistical methods, linear regression provides a straightforward approach to understanding how independent variables, such as daily social media usage, platform type, and engagement metrics (e.g., posts, likes, and comments), influence a dependent variable like dominant emotion. This model allows us to quantify the contribution of each feature to the target outcome through easily interpretable coefficients for our analysis.



Experiments

Dataset

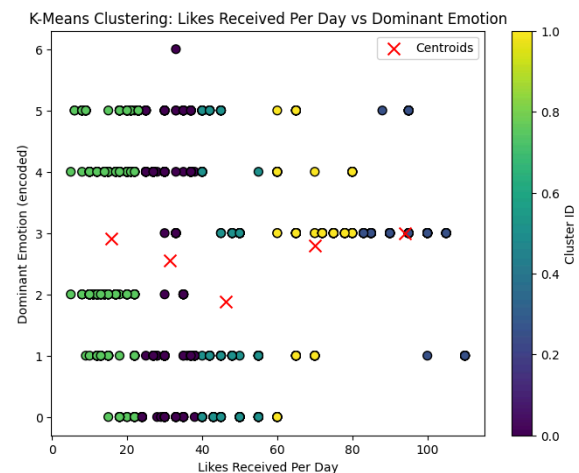
The dataset comprises 1,252 entries and includes various features that capture both behavioral and emotional metrics related to social media usage. It is divided into three subsets: 1,001 entries in the training pool, 103 entries in the testing pool, and 148 entries in the validation pool.

For the linear regression, and Random Forest models, we excluded certain highly correlated

features such as posts per day, likes received, comments received, and messages sent per day. Removing these features prevents overfitting and allows the models to focus on uncovering other meaningful relationships. In the case of linear regression, this step helps avoid multicollinearity, where highly correlated features can distort the regression coefficients. Similarly, for the Random Forest model, removing these features prevents the model from being biased toward them when determining outcomes, resulting in more balanced and accurate predictions.

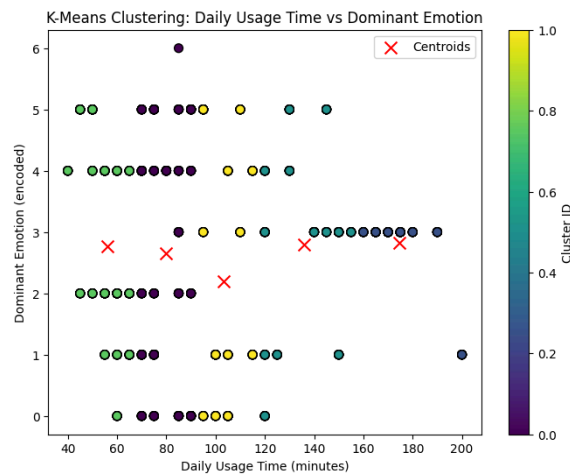
K-Means Clustering

For our clustering we decided to see how certain features affected the user such as likes received daily and their daily usage time. The amount of clusters was set to 5 clusters.



Visualizually seeing the clusters of likes per day and dominant emotion shows that the centroids of each cluster follows a normal gradient where the more negative emotions typically were around the center of approximately 50 likes. It has a silhouette score of 0.54. When looking at daily usage time and dominant emotion it shows

a very similar graph.



The Centroids follow a normal distribution curve with its peak nearing around 100 minutes daily. Although it's dip is not as deep as the likes received per day which shows the lesser importance daily usage time has when it comes to determining someone's dominant emotion. It has a silhouette score of 0.51.

Random Forest

The Random Forest model was trained using the entirety of the cleaned dataset. The primary target was the time spent on social media, while the dominant emotions were one-hot encoded to allow multi-label classification. Handling the encoding of emotions and incorporating platform data posed challenges due to their complexity and potential to confuse the model. Although platform data was included, it was considered for exclusion since it was not the main focus of this study. The model was built using Python's sklearn library RandomForestClassifier, with the number of estimators (decision trees) set to 100, ensuring a robust ensemble learning approach.

The model's evaluation utilized multiple metrics, including the F1 score for precision-recall balance, R^2 to measure variance explained by the predictions, and confusion matrices for visualizing performance across emotion classes.

These metrics provided insights into the model's effectiveness and areas for improvement, highlighting its ability to classify dominant emotions while managing diverse features from the dataset.

FNN

The Feedforward Neural Network (FNN) was designed to predict dominant emotions based on social media usage. The training process began with data preprocessing to prepare the input for the model. Numerical features (e.g., daily usage time, posts per day) were standardized using StandScaler to ensure a uniform scale across features, preventing bias toward any particular variable. Categorical variables (gender, platform) were transformed into binary columns using one-hot encoding, ensuring that these variables were not misinterpreted as ordinal. The target variable, representing dominant emotions, was also one-hot encoded to enable multi-class classification.

The data was split into training and testing sets, with 80% of the data used for training and 20% reserved for testing. Stratification ensured that all emotion classes were proportionally represented in both subsets. During training, 20% of the training data was used as a validation set to monitor the model's performance and prevent overfitting.

The **model architecture** featured an input layer to process the preprocessed features, three hidden layers with 128, 64, and 32 neurons, respectively, each using the ReLU activation function to learn complex non-linear relationships in the data. Dropout layers were added after the first two hidden layers to reduce overfitting. The output layer used softmax activation to generate probabilities for the emotion classes, supporting multi-class classification.

The FNN was trained using **categorical cross-entropy loss** and the Adam optimizer, which dynamically adjusts the learning rate for efficient convergence. Early stopping was implemented to halt training if validation loss did not improve for five consecutive epochs, ensuring the model did not overfit and saving computational resources. The model iteratively updated its weights through backpropagation, minimizing the loss function during each epoch.

After completing the training process, the model was ready for evaluation on the test set to calculate key metrics, including accuracy, precision, recall, and F1 score.

Linear Regression

Linear regression produced an **MSE** of **2.51** and an **R²** of **0.04**, indicating that it explained very little of the variance in the dominant emotion. While features like likes received per day and daily usage time showed slight correlations with emotions, the overall results demonstrated the model's inability to effectively capture the complex relationships within the dataset.

The low r-squared value highlights the limitations of linear regression in modeling non-linear patterns inherent in social media data. Despite its simplicity and interpretability, the model struggled to provide meaningful insights into how social media metrics influence emotional states. These results emphasize the need for more advanced models, such as feedforward neural networks and random forest, to better understand the intricate connections between social media usage and emotions.

Results

The Feedforward Neural Network had the highest accuracy of 99%. The next highest was our Random Forest with an approximate

accuracy of 90%. Our Linear Regression did the worse with only a R^2 of 0.04 and a MSE of approximately 2.51. These results are indicators of low correlation. Finally Our K-Means Clustering has a Silhouette Value of 0.53 at 3 clusters. Which shows that we have poorly defined cluster shapes.

K-Means Clustering

Silhouette Analysis of K-Means Clustering Shows around a 0.53 Silhouette Value between both of the graphs. 0.54 being the highest when dealing with likes received and dominant emotion.

Random Forest

The Random Forest had varying accuracies for each emotion ranging from 94% to the lowest of 83% which can average around to 90%. Random forest also showed inside of its various matrices for each emotion that it was able to predict correctly almost every time, only ending up with one false negative on Anxiety, Boredom, Neutral, and Happiness and false positives on Happiness and Sadness.

R^2 for Anxiety	0.94
R^2 for Boredom	0.92
R^2 for Happiness	0.911
R^2 for Neutral	0.94
R^2 for Sadness	0.83
Average R^2 across all labels: 0.91	

FNN

The neural network training results show a consistent improvement in performance over the course of 100 epochs, achieving 99% on all metrics.

Accuracy	0.99
Precision	0.99
Recall	0.99
F1 Score	0.99

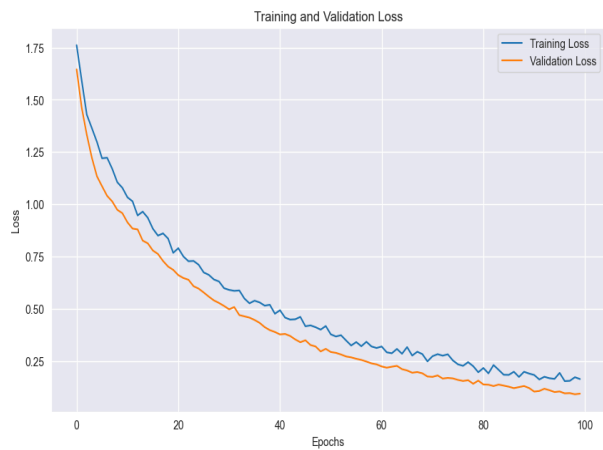
Training and Validation Accuracy:

The graph shows a steady increase in both training and validation accuracy over the epochs, plateauing around 99% accuracy. The close alignment of the two curves indicates that the model generalizes well to unseen data without overfitting. This demonstrates that the model learned the underlying patterns in the dataset effectively.



Training and Validation Loss:

The loss graph depicts a continuous decrease in both training and validation loss over the epochs, with the validation loss remaining consistently lower than the training loss. This suggests that the model not only improved its predictive capabilities but also avoided overfitting, as it maintained high performance on the validation set throughout training.



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Overall Results

Our Random Forest and FNN both yielded accuracies of 90% and above. These values are ideal and can be accepted as a sign that there can be a great correlation between how long someone spends on social media and their emotional state. On the other hand, the Linear

Regression model has gotten considerably lower values. These values can be ignored when considering the overall result of this project. Its poor performance can be attributed to the non-linear trends this data has when comparing features in our dataset.

Conclusion

The best model we tested was the FNN model. It had the highest metrics across the board and beats out all of our other predictive models. Second was the Random Forest model with an average of 0.91 R^2 values. There is still much to improve when it comes to our study as our dataset is very small, only being around 1250 entries. Having more data would allow our models to find out more of the minor impacts other features have on our prediction. The results of our Linear Regression model might even improve due to the increase of data variety we would gain from a larger dataset. This would also allow for our K-means model to have more distinct clusters. Along with that our dataset has around 50% entries that contain “Neutral” for dominant emotion. With a larger dataset it would allow for a potentially more diverse dominant emotion spread. The more diverse emotion spread can help bring forth other features that didn’t impact as much due to the majority neutral emotion spread.

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